

Architecture Justification – Vehicle Parking System

Architecture Selection

We chose Layered Architecture for the Vehicle Parking System.

Architectural Choice

The Vehicle Parking System is designed using a **Layered Architecture** consisting of three main layers: **Presentation Layer, Business Logic Layer, and Data Layer**. The Presentation Layer handles user and administrator interactions, the Business Logic Layer manages parking operations and rules, and the Data Layer stores vehicle, parking slot, reservation, payment, and user information.

Reason 1 – Separation of Concerns

Layered Architecture provides a clear separation between the user interface, business logic, and data management. This is suitable for the Vehicle Parking System because functions such as vehicle registration, parking slot allocation, fee calculation, payment, and parking history can be managed independently within the Business Logic Layer. Changes to the user interface or database can therefore be made without significantly affecting the other layers.

Reason 2 – Centralized Data Management

The system requires consistent management of parking slots, vehicles, reservations, payments, and parking history. A dedicated Data Layer provides centralized access to this information and helps maintain consistency across different system operations. This is particularly useful for preventing incorrect slot allocation and maintaining accurate parking records.

Security Advantage

Layered Architecture allows security controls to be applied at different levels of the system. User authentication and authorization can be handled through the Presentation and Business Logic Layers, while direct access to sensitive data can be restricted to the Data Layer. This helps prevent unauthorized users from accessing administrative functions or protected system data.

Performance Benefit

Layered Architecture provides direct and structured communication between the application layers without the additional network communication required between multiple independently deployed services. This can provide efficient processing for common operations such as checking slot availability, recording vehicle entry and exit, calculating parking fees, and retrieving parking history.